$$(\mathbf{XP} - \mathbf{PX}) = [\mathbf{X}, \mathbf{P}] = i\hbar.$$

Thus, the expression for our factored Hamiltonian becomes

$$\frac{1}{2}(\mathbf{P}^2 + \omega^2\mathbf{X}^2) + \frac{1}{2}i\omega i\hbar$$

or

$$\frac{1}{2}(\mathbf{P}^2 + \omega^2\mathbf{X}^2) - \frac{1}{2}\omega\hbar.$$

In other words, the factored expression we started out with in Eq. 10.19 is actually smaller than the Hamiltonian by $\frac{\omega\hbar}{2}$. To recover the actual Hamiltonian, we need to add the $\frac{\omega\hbar}{2}$ back in:

$$\mathbf{H} = \frac{1}{2}(\mathbf{P} + i\omega\mathbf{X})(\mathbf{P} - i\omega\mathbf{X}) + \frac{\omega\hbar}{2}.$$

Rewriting the Hamiltonian this way and that way may seem like an exercise in futility, but trust me, it's not. First of all, the last term is just an additive constant that adds the numerical value $\frac{\omega\hbar}{2}$ to every energy eigenvalue. We can ignore it for now. Later, after we've solved the rest of the problem, we can add it back in. The guts of the problem are found in the expression (**P** + *i*ω**X**) (**P** − *i*ω**X**). It turns out that these two factors, (**P** + *i*ω**X**) and (**P** − *i*ω**X**), have some very remarkable properties. In fact, they are the raising and lowering operators (or creation and annihilation operators) that I told you about earlier. For now, these are just names, but as we go along we'll see that the names were well chosen. The obvious definitions would be

$$\mathbf{a}^- = (\mathbf{P} - i\omega\mathbf{X})$$